

ROSHAN RAJ

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science

Sept 2023 - Dec 2026

- GPA: 3.732, **Dean's Honor List**
- Research Assistant on the TIPPERS project led by Dr. Sharad Mehrotra.
- Technology Intern at **Venture Capital Society @ UCI** - developed and maintained the club's website
- Core Member at **Google Developer Student Club (GDSC)** — contributed to dev projects and hosted technical workshops.
- **Relevant Coursework:** *Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Software Engineering, Computer Architecture, Software Testing & Quality Assurance, Machine Learning, Artificial Intelligence, Computer Architecture & Databases, Distributed Systems, Information Retrieval, Formal Languages & Automata, Discrete Mathematics*

EXPERIENCE

TheCoderSchool ☞

Irvine, CA

Instructional Software Developer

Dec 2024 - Mar 2025

- Developed and delivered **software engineering curriculum** in data structures, OOP, and algorithm design, equipping learners with industry-relevant skills and improving task efficiency by 20%.
- Integrated testing frameworks and API-based validation tools into Python and C++ projects, reducing error rates by 25% and improving overall deployment quality.

Dreams for Schools ☞

Santa Ana, CA

Software Engineering Instructor

Aug 2024 - Dec 2024

- Designed interactive computer science curriculum in **Python, C++, HTML/CSS, and Java** with focus on data handling, loops, and control structures, achieving a 95% learner satisfaction rate.
- Directed 15+ project-based coding initiatives applying TDD, pair programming, and algorithmic thinking, resulting in a 98% student success rate on assessments.

TIPPERS (T-Board) @ UCI ☞

Irvine, CA

Research Assistant

Sept 2023 - Dec 2024

- Enhanced IoT analytics platform using **C++** and **OpenGL**, optimizing data structures and queries to improve real-time processing speed by 40% for interactive decision-making.
- Implemented **Docker**-based deployments for scalable, privacy-focused simulations, supporting automated data ingestion pipelines.
- Developed occupancy monitoring tools for 50+ rooms, enabling insights that cut energy consumption by 15% during low usage.

PROJECTS

Sentinance ☞

- Constructed a real-time financial sentiment platform using **Python, FastAPI, FinBERT, and PRAW**; decreased data latency by 15% using optimized modular data pipelines for speedy financial insights.
- Assembled interactive dashboards in **React, Vite, and Recharts** with live sentiment timelines, leveraging real-time API integration and scalable backend architecture designed for multi-source data ingestion.

SkillSync ☞

- Engineered the development of SkillSync, an AI-driven career platform using **FastAPI, spaCy, and Groq LLMs**, enabling resume parsing with 95% accuracy and generating over 200 mock interviews.
- Produced 4 full-stack dashboards using ReactJS, Node.js, Next.js, and Recharts to visualize skill progression by leveraging LeetCode/GitHub APIs, reducing query response time by 22%.

InsightIQ ☞

- Authored an AI-powered analytics dashboard using **FastAPI, Groq, and LangChain** for NL-to-SQL query generation and result explanations, integrating PostgreSQL/DuckDB with schema creation from CSV uploads.
- Devised multi-user dashboards with **Next.js, Chakra UI, and Recharts**, featuring drag-and-drop widgets, fine-grained access control, and Dockerized deployments with **Supabase and AWS S3**.

Resume Analyzer ☞

- Launched an AI-driven resume analysis platform using **FastAPI, Uvicorn, PyPDF2, and Groq LLMs**, applying NLP techniques to extract key insights and boost analysis accuracy by 30%.
- Assembled an interactive frontend with **JavaScript and modern frameworks**, integrating feedback loops to identify skill gaps and optimize resumes for ATS compatibility and job targeting.

Search Engine ☞

- Pioneered a modular search engine with **Python**, leveraging **NLTK** to categorize 10,000+ documents; fine-tuned PageRank/TF-IDF algorithms, achieving query results in under 50ms and a 10% better recall rate.
- Conceived end-to-end ranking and retrieval pipelines using NLTK and Python, enabling 95%+ query coverage on benchmark datasets, while supporting scalable extensions for complex search features.

Fashion MNIST Classifier ☞

- Trained and evaluated four machine learning models — **k-NN, Logistic Regression, Random Forest, and a PyTorch Neural Network** — on the Fashion-MNIST dataset.
- Implemented standardized training pipelines, data preprocessing, and model evaluation using **scikit-learn and PyTorch**, generating classification reports and confusion matrices. Conducted hyperparameter tuning and error-pattern analysis to optimize accuracy and assess classifier robustness.

DeepOutpaint ☞

- Engineered a GAN-based image outpainting system using **DCGANs**, extending images beyond visible boundaries with contextual realism and achieving 90% accuracy in user testing.
- Integrated pixel-level blending and auto-encoder refinement techniques to enhance visual continuity and aesthetic quality across diverse image inputs.

SKILLS SUMMARY

- **Languages:** Python, C++, C, Java, C#, JavaScript, HTML, CSS, SQL, R, Bash, Spring Boot, XML, JSON
- **Frameworks:** ReactJS, Next.js, Node.js, Blazor, FastAPI, LangChain, Flask, Chakra UI, Agile, spaCy, PL/pgSQL, Scrum
- **Tools:** Git, Docker, PostgreSQL, MySQL, GitHub Actions, Xcode, API Integration, Uvicorn, PyPDF2, RStudio
- **Platforms:** Linux, macOS, Windows, Web, AWS, Google Cloud Platform (GCP), Arduino, .NET